Tactile FLORA: Fingertip Force as Text Improves a Vision-Language Reward Model's Grasp Judgments

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Abstract

Vision-language reward models judge whether a robot completed a task from video alone, yet the outcome of a grasp is often decided at the fingertips, where a camera cannot see whether contact was made, held, or lost. We add a fingertip-force channel to NVIDIA's FLORA-Rewarder (a Qwen3.5-27B video reward model) without changing its architecture: the force stream is normalized, rendered as a short text block with a label-free summary line, and placed in the prompt after the video. Fine-tuning with LoRA on 350 grasp clips from the HRDexDB Inspire-hand dataset, we compare the tactile model with the released model and with a matched fine-tune on the same clips and settings but with the force block omitted. On 251 held-out clips over 32 unseen objects the tactile model reaches 79.3% accuracy and 0.873 AUROC against 76.5% / 0.830 for the video-only fine-tune and 66.5% / 0.600 for the released model; the ranking gain over the matched control is $+0.043$ AUROC (object-level bootstrap 95% interval $+0.001$ to $+0.080$, $p \approx 0.02$). Shuffling the force blocks between clips removes the gain ($p = 0.009$) and blanking the video leaves only the force's own moderate signal (AUROC 0.775), so the model reads both channels. The gain concentrates on the clips where the video-only model is unsure. On ten live rollouts of an ACT policy on a Rethink Baxter with home-built force-sensing fingertips, evaluated zero-shot, the force block takes the ranking of the two successful grasps from below chance (AUROC 0.31 for the video-only fine-tune, 0.38 for the released model) to perfect (1.0), while shuffled force gives 0.25 and removed force 0.44. Force rendered as text is a cheap, inspectable way to give a video reward model the contact information it lacks.

1 Introduction

Reward models that watch a video and score whether a manipulation task succeeded are becoming the labelling engine behind robot learning at scale: they rank rollouts, filter demonstrations and supply rewards for policy improvement. FLORA Leng et al. [2026] is a recent example: a family of fine-tuned Qwen3.5 vision-language models whose reward member takes a multi-camera clip and a task instruction and answers with an outcome score from 1 to 5. It is vision-only.

Grasping is the case where vision-only judgment is weakest. The decisive events of a grasp, whether the fingers actually closed on the object, how firmly they hold it, whether the hold is slipping and when contact ends, happen at the fingertips, occluded by the hand itself, and leave few pixels in a camera view. A loaded gripper lifting an object and an empty gripper lifting nothing can look

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identical from a head camera. Fingertip force sensors see exactly these events, and they are cheap. Our hypothesis is that *fingertip force reveals contact failures that RGB misses or sees late*, and that a video reward model can be taught to use it.

The question is how to give a large vision-language model a new sensor without retraining it. Learned encoders and adapter tokens are possible but opaque and data-hungry. We take the simplest route: render the force as text. The force stream is resampled to the frames the model already samples, normalized so that 0 is free space and 1 is a typical grasp, and written as one line per frame with a label-free summary line at the top. The block enters the prompt after the video, with one sentence explaining its columns. The model is then fine-tuned with LoRA on outcome-labelled clips. Everything else, the released weights, prompts, frame sampling and answer format, stays as released.

Our contributions are:

- A force-as-text representation for a video reward model, with the normalization, summary statistics and a small force-only classifier probability that make the signal legible to the language model (Section 3).
- A controlled evaluation on object-disjoint held-out splits of HRDexDB Lim et al. [2026], with two controls that isolate the force contribution: a matched fine-tune without force, and a shuffled-force ablation (Section 6).
- A zero-shot transfer test on a different embodiment: a Rethink Baxter with force-sensing fingertips we built, running a learned policy, where the force block decides the ranking of successes over failures (Section 6.2).
- An account of what went wrong first. A naive force rendering produced no gain; an audit of the block the model actually read found it nearly uninformative, and correcting the representation, not the model, produced the result (Section 5).

2 Related work

Vision-language reward models. FLORA Leng et al. [2026] fine-tunes Qwen3.5 models into a segmenter, a captioner and a reward model that scores task outcome and dense execution quality from video; related video reward models and benchmarks include RoboReward Lee et al. [2026] and Robometer Liang et al. [2026], and vision-language models have been used as success detectors since SuccessVQA Du et al. [2023]. These models are vision-only. We keep FLORA's reward model and add one sensor channel to its prompt.

Tactile sensing for grasp outcome. Fingertip force has long been used to detect grasp success, slip and contact loss Veiga et al. [2015], Yuan et al. [2017], Calandra et al. [2017], and recent datasets pair tactile streams with manipulation video Lim et al. [2026]. Our force-only probes agree with that literature: force statistics alone separate HRDexDB grasp outcomes at AUROC 0.76 to 0.80, a real but moderate signal, weaker than video alone (0.90).

Touch-language models. TVL Fu et al. [2024] aligns a tactile encoder with vision and language on 44K GelSight-style image pairs; Octopi Yu et al. [2024] trains a tactile-language model that reasons about object properties from GelSight videos; and UniTouch Yang et al. [2024] binds several vision-based touch sensors to a shared embedding space. All three learn a tactile encoder from high-resolution tactile images. Our signal is the opposite case, a handful of low-dimensional force channels, and we make no architectural change: the force is serialized as text and read through the model's existing tokenizer.

Sensor data as text for language models. Serializing time series and sensor readings as text for a language model is a known technique Gruver et al. [2023], Jin et al. [2024], Xu et al. [2024]. Our contribution is not the idea but the demonstration that a video reward model can learn to combine such a block with video under LoRA, and the controls that show it does.

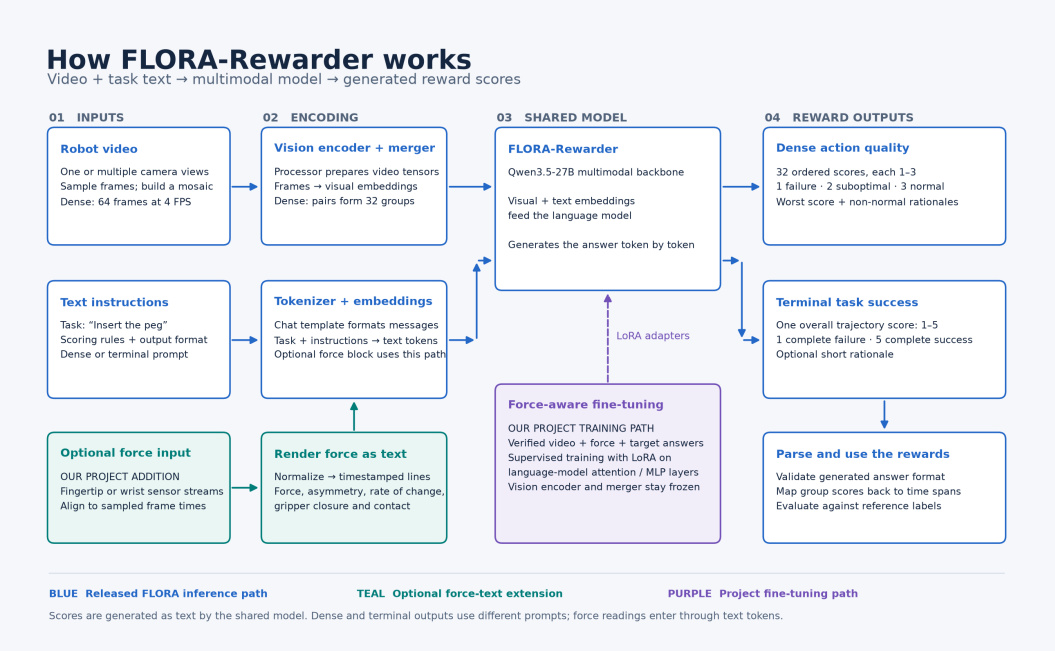


Figure 1: How the force channel enters FLORA-Rewarder. The released inference path (blue) takes the video mosaic and the task text; our addition (teal) normalizes the force stream, renders it as timestamped text lines and feeds them through the same tokenizer; LoRA adapters on the language model (purple) are the only trained weights. No architectural change is made.

3 Method

3.1 Model

FLORA-Rewarder Leng et al. [2026] is a Qwen3.5-27B Qwen Team [2026] vision-language model (Figure 1). For the outcome task it receives up to 64 frames sampled uniformly from the clip, composed into a mosaic of up to four camera views, and a task instruction, and answers with a score from 1 (failure) to 5 (success). We use the released weights and prompts, one scene camera, the released frame sampling, and 96-pixel mosaic cells. The continuous score we evaluate is the expected score $\mathbb{E}[s]$ over the digits 1 to 5 at the answer token, with 4.0 as the success threshold, as in the released evaluation.

3.2 The force block

The raw sensor stream (50 Hz per-finger force for the Inspire hand, 200 Hz pad counts for Baxter) is resampled to the 64 sampled frame times, normalized per source so that 0 is free space and 1 is the median grasp force of the training set, and rendered as one line per frame plus a summary line. Per-pad resting offsets are tared per trial (the median of the first second is subtracted and the result floored at zero), channels are ranked by peak excess, and the two most loaded pads are shown. A one-sentence explanation of the columns precedes the video. An abbreviated block for one HRDexDB training clip:

```
Force (normalized: 0 free space, 1 median grasp, max 3):
summary R: contacts=1 longest=6.00s total=6.00s peak=1.83 hold=5.67s holdF=1.55 late/early=0.70 low=0%
     force-only classifier P(success)=0.76
t=0.00 L none | pad R F=0.00 D=0.00 dF=0.00 g=0.00 c=0
t=0.25 L none | pad R F=0.00 D=0.00 dF=0.00 g=0.00 c=0
...
t=12.50 L none | pad R F=1.62 D=0.21 dF=0.03 g=0.71 c=1
```

F is total grip force, D the pad asymmetry, dF its rate, g the hand closure (0 open, 1 closed) and c a contact flag (force above a threshold set at 5% of the grasp force). The summary line is computed

label-free from the same normalized features: number of contact segments, longest and total contact, peak force, hold duration and hold force, the late-over-early hold ratio, and the fraction of low-force hold. It ends with the success probability of a small gradient-boosted classifier (120 trees of depth 2) over 31 trial-level force statistics, fitted on the 350 training clips only. Training clips see the out-of-fold probability of a fold model that never saw them, so the language model learns how far to trust the line rather than over-trusting in-sample predictions. A block is capped at 3,000 tokens (largest observed: 2,700). Under the shuffled and removed ablations the summary line is recomputed from whatever block the model is given, so no ablation leaks the clip's own force.

3.3 Fine-tuning

Both adapters use LoRA Hu et al. [2022] with rank 32 and $\alpha = 64$ on all attention and MLP projections of the language model, with the vision encoder frozen. The target is the released outcome answer template with the score digit. Settings, fixed before any held-out result was read: four epochs, seed 17, learning rate 2×10^{-5} with 10% warm-up, effective batch 16, gradient clipping 1.0, 88 optimizer updates, final checkpoint only. The force normalization was refitted on the training clips only (median grasp 1,016 raw units, contact threshold 49.4). Training ran on two AMD MI210 GPUs.

3.4 Controls and ablations

Every claim is backed by two controls on identical data:

- **Matched no-force fine-tune** ("fine-tuned FLORA"). Same base, clips, seed, epochs, learning rate, batch and LoRA, with the force block omitted. A gain over it is a force gain, not a fine-tuning gain.
- **Shuffled force.** The tactile model evaluated with each clip's block replaced by another clip's. A gain that survives shuffling is not a force gain.

Two diagnostics complete the picture: **force removed** (the tactile adapter with no block) and **video blanked** (frames mid-gray, force only), which show whether the model reads the block at all and what the force carries on its own.

4 Experimental setup

4.1 HRDexDB

HRDexDB Lim et al. [2026] records a 12-DoF Inspire DFTP hand on a UR arm grasping household objects in a 23-camera rig, with per-finger force at 50 Hz (six channels; the two most loaded are rendered) and a success flag per trial (object lifted and held for 10 s). We use one fixed scene camera (id 22641005) and the released labels. Clips run 12.5 to 74.6 s (median 27.5 s), so every clip receives the full 64 uniformly sampled frames of the released outcome sampling. All 618 Inspire DFTP trials were downloaded at a pinned revision with the recorded camera timestamps, and sensor time was mapped to video time through them (this corrects both the start offset of 2.2 to 2.3 s and the clock-rate difference between the two streams). Objects were frozen into object-disjoint splits before any model saw them: the pilot's train and validation objects kept their assignment, and the remaining objects were hash-ordered with the seed string `hrdexdb-v2:20261021` to reserve 16 objects for test and eight more for development; after 17 mechanical exclusions (invalid arrays, conflicting labels, undecodable video, duplicate videos), the splits are **350 train / 136 development / 115 test clips over 52 / 16 / 16 objects**. Test clips were scored once, with settings fixed on the development set.

4.2 Baxter

To test transfer to our own hardware we built a tactile capture stack on a Rethink Baxter.

Sensor. Four thin-film force-sensing resistors (two per gripper) in a 3.3 V divider with a 4.7 kΩ reference, read by a Teensy 4.0 at 200 Hz (12-bit ADC, 16-sample oversampling), published into ROS on the robot clock with a linear clock fit refreshed every 5 s. Only the left gripper's pads were mounted for these rollouts. Counts are uncalibrated (no weight calibration file was present), with a

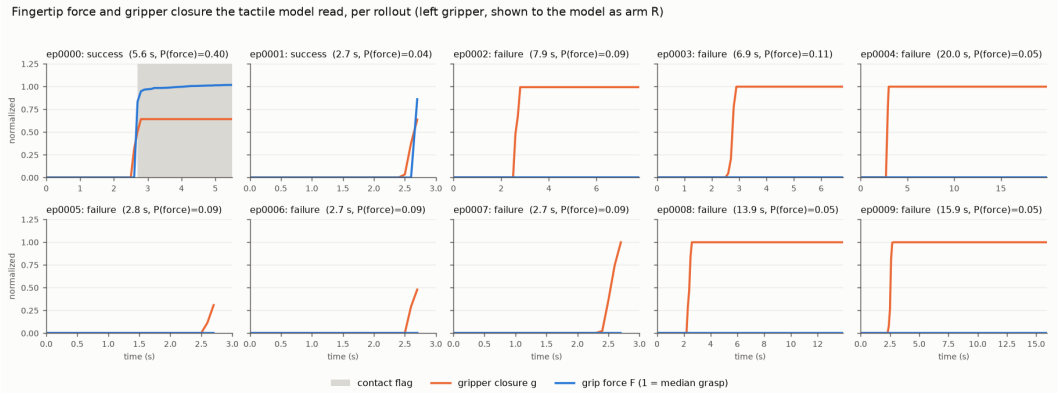


Figure 2: Normalized fingertip force and gripper closure for the ten Baxter rollouts. Only the two successes (ep0000, ep0001) show contact; ep0001's contact is a single sampled frame at the very end. The eight failures close fully on nothing (closure 1.0) and several then lift the empty gripper.

resting offset of 92 counts per pad; the same normalization pipeline handles them, fitted label-free on the ten rollouts (baseline 184 counts for the two pads, median grasp 892.5, contact threshold 219). Gripper closure is derived from Baxter's gripper position (0 to 100, scaled to 0 to 1, open about 0.96) as $(0.96 - \text{position})/0.92$, so a close stopped by the 40 mm cube reads 0.64 and an empty close 1.0.

Demonstrations and policy. Fifty hand-guided left-arm cube picks ("pick up the red cube") were recorded at 15 fps from a USB webcam mounted on Baxter's head and the left wrist camera, both at 640×360, with joint state and gripper position (no tactile data): the operator squeezes the cuff to put the arm in zero-gravity mode, guides it to the cube, closes the gripper with the cuff button and lifts. Each demo is cut at the first sign of the gripper re-opening after its grasp and published as a LeRobot Cadene et al. [2026] (v0.6.1) dataset. An ACT policy Zhao et al. [2023] (chunk size 30, 15 action steps, ResNet-18 backbone, VAE, learning rate 10^{-5}, inputs: the 8-dimensional state plus both cameras) was trained for 50,000 steps on five RTX 2080 Ti GPUs (global batch 65, fp16) with episodes 45 to 49 held out; held-out action L1 fell from 0.0325 at step 5,000 to 0.0267 at step 50,000. The rollouts used the step-30,000 checkpoint (held-out L1 0.0278).

Rollouts. Ten live rollouts of the policy, commands at 30 Hz interpolated between the 15 Hz action rows, a 1 s replan, a 20 s limit and safety checks (a joint-jump guard of 0.25 rad between the commanded chunk and the measured arm, a fingertip box, a speed cap), recorded with the same cameras plus the 200 Hz tactile stream, which the policy never reads. The operator labelled episodes 0 and 1 successes and the other eight failures. Episode 0 is a clean grasp: the gripper stops at closure 0.64 on the cube, force rises to the grasp level and holds for 2.9 s while the arm lifts the cube 20 cm. Episode 1 was stopped by the joint-jump guard at 2.7 s with the cube between the pads and force appearing 0.03 s before the last frame. The eight failures closed on nothing (closure 1.0, zero force); five of them then lift and move the empty gripper exactly as a success would (Figure 2).

The three HRDexDB models were evaluated on these ten clips zero-shot: no Baxter clip was used for training, normalization constants were fitted on the ten rollouts (label-free), and the left gripper's two pads were presented as the "pad R" lines the adapters had seen in training.

4.3 Metrics

Accuracy is the share of clips called correctly at the 4.0 threshold on $\mathbb{E}[s]$. AUROC is the probability that a random success is scored above a random failure (ties count half). MAE is the mean absolute error of the *integer* hard score against the truth score (5 for a success, 1 for a failure) on the 1-to-5 scale, multiplied by 100 following the FLORA evaluation code; it therefore ranges from 0 to 400, not 0 to 100. A model that answers "2" on every clip of a set with 20% successes scores $100 \times (0.2 \cdot 3 + 0.8 \cdot 1) = 140$, which is the released model's Baxter value; answering "1" everywhere scores 80. For HRDexDB, paired differences are tested with an object-level bootstrap (AUROC) and

Table 1: HRDexDB held-out results. Accuracy at the 4.0 threshold, AUROC, MAE of the hard score ($100\times$ mean $|\text{score} - \text{truth}|$ on the 1–5 scale, range 0–400), and recall per class. The two fine-tuned models differ only in the force block.

Split	Model	Acc.	AUROC	MAE	Succ. rec.	Fail. rec.
dev (136)	released, video only	66.2%	0.614	150.0	0.878	0.403
	released, video + force	66.2%	0.648	148.5		
	fine-tuned, video only	82.4%	0.899	70.6	0.784	0.871
	fine-tuned tactile	**85.3%**	**0.926**	**52.9**	0.838	0.871
	tactile, shuffled force	73.5%	0.824	108.8		
	tactile, video blanked	53.7%	0.775	120.6		
test (115)	released, video only	67.0%	0.593	143.5	0.768	0.522
	released, video + force	70.4%	0.647	134.8		
	fine-tuned, video only	69.6%	0.733	121.7	0.725	0.652
	fine-tuned tactile	**72.2%**	**0.800**	**97.4**	0.797	0.609
pooled (251)	majority class	57.0%	0.500			
	released, video only	66.5%	0.600			
	fine-tuned, video only	76.5%	0.830			
	fine-tuned tactile	**79.3%**	**0.873**			

the exact McNemar test (accuracy). On Baxter, with two successes and eight failures, AUROC is a count over 16 ordered pairs and no significance test is attempted.

5 Experiment process

The result did not come from the first attempt, and the path matters for anyone repeating it.

Pilot (32 train / 16 validation clips). The first LoRA runs showed no force benefit. An audit found the real limits: a tiny sample, labels verified only for plumbing, and force and video clocks that had never been aligned.

Expanded data, naive force rendering. With all 618 clips, aligned clocks and frozen object-disjoint splits, both adapters were trained with the recipe above. On the 136 development clips: video-only 82.4% / AUROC 0.899, force 81.6% / 0.902, shuffled force 78.7% / 0.885. Still no force benefit. A deep dive into the block the model had actually read found three defects in the representation: the hand-closure sign was inverted (Inspire joint angles run 1000 open to 0 closed), so normalization had fitted "grasp" on open-hand frames and the scale was 37.5 raw units against contact peaks of 1,000 to 3,000, saturating every contact at the clip maximum; per-trial resting offsets exceeded that scale, so one clip started at maximum force with the hand open; and the contact flag fired on every nonzero reading. A classifier over the rendered lines reached only AUROC 0.57, against 0.78 on the raw force with a per-trial tare. The block was nearly uninformative, which is why the model ignored it.

Corrected representation (the model reported here). Only the representation changed: per-trial tare, channels ranked by peak excess, sign-corrected closure, a 5% contact floor, the summary line and the classifier probability. Splits, window ids, base, recipe and seed stayed fixed. The force arm finished its 88 updates (loss 2.49 to 0.02); the matched control arm of this run was preempted by the cluster scheduler, so the control adapter from the previous run, whose inputs are identical, serves as the video-only comparison. The only unmeasured difference is run-to-run training noise.

6 Results

6.1 HRDexDB

Table 1 and Figure 3 give the held-out numbers. Against the released model, the tactile fine-tune adds 12.8 accuracy points and 0.27 AUROC over the 251 pooled clips. Against the matched video-only fine-tune, it adds 2.8 points and 0.043 AUROC, with lower MAE on both splits (52.9 against 70.6 on

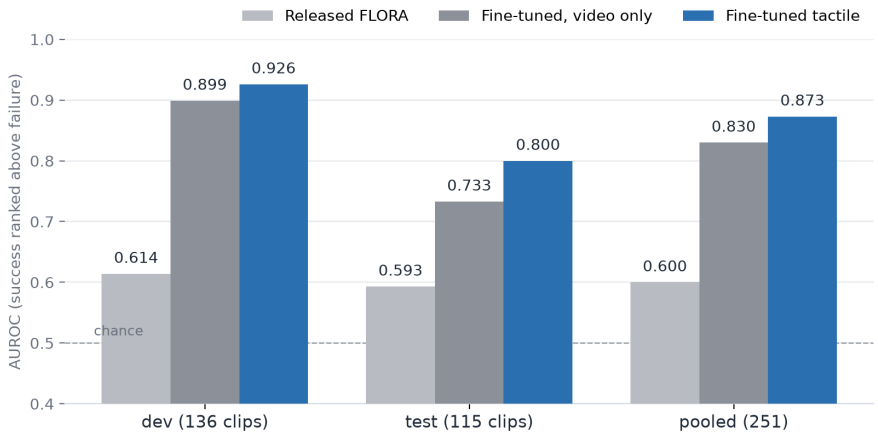


Figure 3: HRDexDB held-out AUROC of the released model, the matched video-only fine-tune and the tactile fine-tune on the development, test and pooled clips. The dashed line is chance.

Table 2: Where the force helps. Clips are split by how sure the video-only fine-tune is: within 0.75 of the 4.0 threshold on $\mathbb{E}[s]$ (uncertain) or further (confident). Pooled over both splits.

Clips	n	Video-only accuracy	Tactile accuracy
uncertain (video-only $\mathbb{E}[s]$ within 0.75 of threshold)	101	66.3%	**72.3%**
confident	150	83.3%	84.0%

development, 97.4 against 121.7 on test). The tactile model's extra correct calls are mostly successes: its success recall is higher on both splits while failure recall stays level.

Statistics. The AUROC gain over the video-only fine-tune is $+0.043$ pooled over 251 clips, object-level bootstrap 95% interval $+0.001$ to $+0.080$, $p \approx 0.02$; per split it is $+0.026$ on development (not significant) and $+0.067$ on test ($p \approx 0.04$). The accuracy gain is not significant on its own: the tactile model fixes 26 clips the video-only model gets wrong and breaks 19 it gets right (12 against 8 on development, 14 against 11 on test; exact McNemar $p = 0.37$).

The model reads the force. With shuffled blocks the tactile model drops from 85.3% to 73.5% on development and loses 25 paired clips while winning 9 ($p = 0.009$), landing below the video-only fine-tune. With the video blanked it keeps AUROC 0.775 but predicts failure almost everywhere (accuracy 53.7%): that is the force's own signal, consistent with the force-only probes (0.76 to 0.80). Neither channel alone explains the result; the combination does.

Where the force helps. Table 2 splits the clips by the video-only model's own confidence. On the 101 clips where it sits within 0.75 of the threshold, the tactile model is 6 points more accurate; on the 150 confident clips, under 1 point (on development alone the uncertain band is 46 clips at 71.7% against 82.6%). The force settles the calls the video cannot make and leaves the confident ones alone.

What the wins look like. Of the 26 clips where the tactile model is right and the video-only model wrong, the pattern is consistent: the video-only model under-calls successes whose hold is visually unremarkable (long, steady contact above the median force) and over-calls failures where the hand closes convincingly on camera but the force shows a weak grip, a re-grasp, or contact that ends early. The three largest margins are in Table 3; Figure 4 shows the first. The 19 clips the video-only model wins are mostly successes the tactile model scored 3.1 to 3.8, just under the threshold, and failures with long, strong contact that force cannot distinguish from a hold.

A stacking check. A logistic stack of the video-only score, the tactile score and the force-only classifier probability, fitted object-grouped on development, reaches 88.2% / 0.928 on development and 76.5% / 0.800 on test; the force classifier alone gets 72.8% / 0.787 and 76.5% / 0.777. On the

Table 3: The three held-out clips with the largest margin in favour of the tactile model. Scores are $\mathbb{E}[s]$ out of 5; success at 4.0 or above. Force columns are read from the block the tactile model saw.

Clip	Truth	Video-only	Tactile	What the force showed
standing frame #4 (test)	success	1.27	**4.19**	peak 2.5× grasp, 16.4 s contact ending 4.5 s before clip end
serving bowl #3 (dev)	failure	4.66	**2.29**	1.4× grasp but only 6.7 s of contact in a 39 s clip
cactus #6 (dev)	failure	4.19	**1.98**	1.8× grasp, contact ends at 14.7 s of 20.2 s

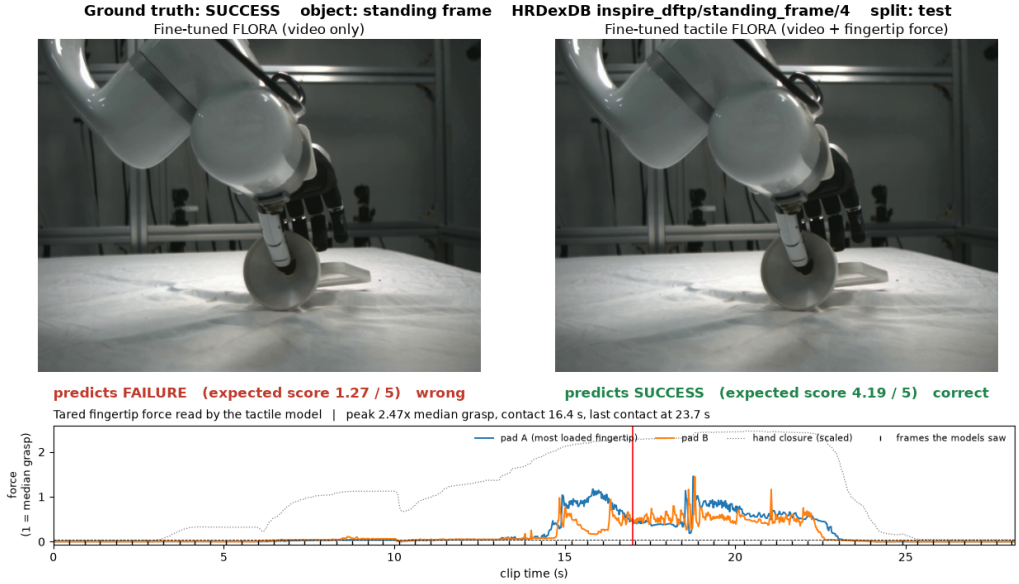


Figure 4: Frame from the side-by-side video of HRDexDB test clip standing frame #4 (success): the scene camera with each model's prediction, above the tared fingertip force the tactile model read (two most loaded pads, hand closure, the 64 sampled frames as ticks). The video-only fine-tune scores 1.27; the tactile model, reading 16.4 s of contact at 2.5× the median grasp force, scores 4.19.

harder test objects the small force classifier is competitive with both language models on accuracy, so the tactile model does not yet fully exploit the line it is given: the language model, not the sensor, is the bottleneck.

6.2 Baxter rollouts

Tables 4 and 5 and Figure 5 give the six conditions. Every accuracy is the majority-class 0.80: no model scores either success at or above 4.0, so the hard answers carry nothing and the comparison lives in the ranking. There the two conditions that see the clip's own force block rank both successes above all eight failures (AUROC 1.0), and the four that do not rank below chance.

- **ep0000, the clean grasp.** The tactile model ranks it first of ten ($\mathbb{E}[s]$ 2.41). The video-only fine-tune ranks it sixth, below five failures; the released model eighth, and its rationale says the cube "remained on the table in the final frames" while the cube was 20 cm in the air (Figure 6). At 96 pixels the video models do not see the lift; the force block does.
- **ep0001, one contact frame at the end.** The tactile model ranks it second, above every failure; video-only ranks it eighth. One contact line with closure 0.64 (stopped by the cube) rather than 1.0 (closed on nothing) was enough.
- **ep0003 and ep0008, long empty lifts.** The video-only model's two highest scores (1.99, 1.75): the arm closes, lifts and moves exactly as in a success. The tactile model gives them 1.09 and 1.04; with its force block removed they jump to 2.97 and 2.31, the top of that column.

Table 4: Six conditions on the ten Baxter rollouts (two successes). Accuracy at 4.0, AUROC over the 2×8 success–failure pairs, MAE of the hard score ($100\times$ mean $|\text{score} - \text{truth}|$, range 0–400), and the hard scores each condition emitted.

Condition	Acc.	AUROC	MAE	Hard scores given
released FLORA, no force	0.80	0.375	140	2 on all 10
released FLORA + force block	0.80	**1.000**	100	1 on 4, 2 on 6
fine-tuned FLORA, video only	0.80	0.313	80	1 on all 10
fine-tuned tactile FLORA, real force	0.80	**1.000**	80	1 on all 10
tactile, shuffled force	0.80	0.250	90	1 on 9, 2 on 1
tactile, force removed	0.80	0.438	80	1 on all 10
force-only classifier (printed in the block)	0.80	0.500		

Table 5: Expected score $\mathbb{E}[s]$ per Baxter rollout and condition. No cell reaches the 4.0 threshold; bold marks a success scored above every failure in its column. Only the two force-seeing conditions do that.

Episode	Label	Released	Rel. + force	Video-only	Tactile	Tact., shuffled	Tact., no force
ep0000	success	1.91	**2.41**	1.34	**2.41**	1.03	1.43
ep0001	success	2.00	**1.92**	1.05	**1.60**	1.03	1.12
ep0002	failure	2.01	1.78	1.40	1.05	1.05	1.52
ep0003	failure	2.03	1.81	1.99	1.09	1.08	2.97
ep0004	failure	1.91	1.06	1.60	1.02	1.02	1.32
ep0005	failure	1.82	1.05	1.05	1.03	1.03	1.05
ep0006	failure	1.99	1.00	1.05	1.02	1.62	1.11
ep0007	failure	1.85	1.48	1.03	1.03	1.03	1.06
ep0008	failure	2.04	1.70	1.75	1.04	1.06	2.31
ep0009	failure	2.02	1.74	1.55	1.07	1.07	2.02

The tactile model's lowest success (1.60) sits 0.51 above its highest failure (1.09); the released model with the force block also separates the classes, but by 0.11. Shuffling the blocks drops the tactile AUROC to 0.25 and ep0000 from 2.41 to 1.03; removing them gives 0.44 with the empty lifts on top. The force-only classifier printed in the block scores 0.5 on these clips (it ranks ep0001, with its 0.13 s hold, last), so the robot result cannot come from that line alone. The ranking is carried by the model reading the force text, and the direction matches HRDexDB: force helps where the video is ambiguous, and here the video is ambiguous on every clip.

7 Limitations

Sample size. 251 held-out clips over 32 objects cannot certify a 3-point accuracy gain (about 1,000 clips would be needed at the observed win/loss ratio); the AUROC gain is the significant one. The Baxter set has ten clips and two positives: a perfect AUROC there is 16 of 16 ordered pairs, and no significance claim is made.

Control provenance. The video-only comparison is the control adapter of the previous training run; the matched control of the final run was preempted. Its inputs are identical, so the only unmeasured difference is run-to-run training noise, which this experiment cannot report.

Part of the gain rides a small classifier. The block's summary line ends with the probability of a gradient-boosted force classifier, and the language model reads that line as well as the raw force lines. The frame lines and plain statistics remain in the block, and the Baxter result cannot come from the classifier, whose own AUROC there is 0.5.

Undertrained. Training loss reaches its floor by update 20 of 88 because the target is a constant template plus one digit; both fine-tuned models are data-limited.

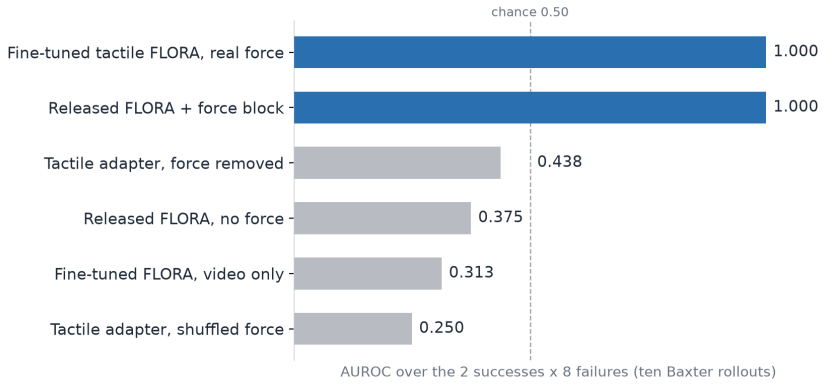


Figure 5: AUROC of the six conditions on the ten Baxter rollouts (two successes, eight failures). Only the two conditions that see the clip's own force block rank both successes above every failure.

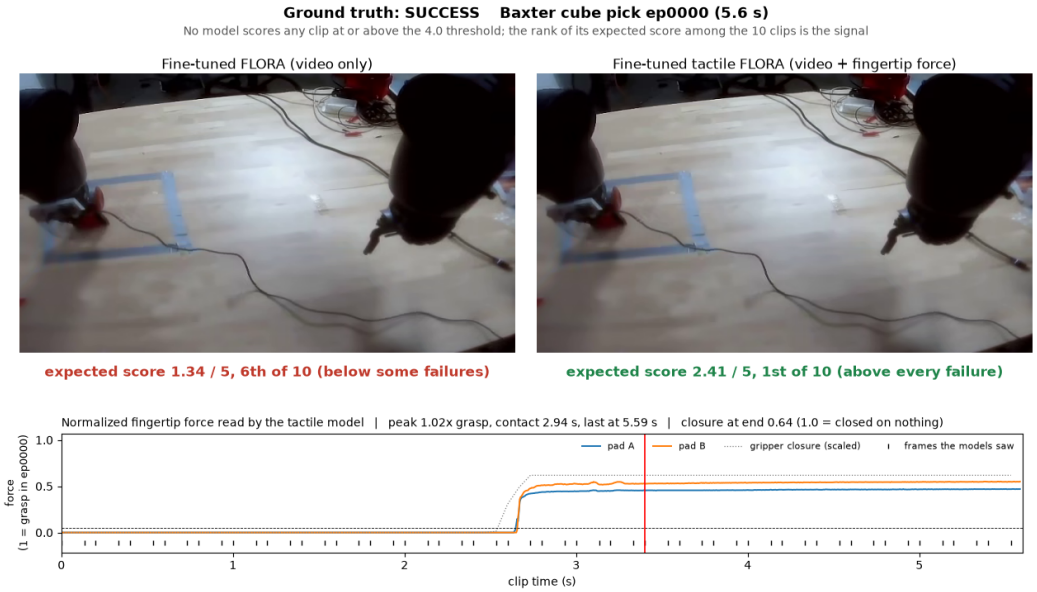


Figure 6: Frame from the side-by-side video of Baxter rollout ep0000 (success): head camera with each model's expected score and rank, above the normalized fingertip force the tactile model read. The video-only fine-tune ranks this clean grasp sixth of ten; the tactile model ranks it first.

Input resolution. Both fine-tuned models and the released baseline saw each camera as a 96-pixel cell. The released model was trained on far larger views; higher-resolution inputs would likely help every video model, and the force gain may shrink as the video gets stronger.

Baxter specifics. Normalization was fitted on the ten test clips (label-free, but the test set); episode 1's success rests on one line of the block; the left gripper was presented as the right-hand pads the adapters knew; the force counts are uncalibrated.

8 Conclusion

Rendering fingertip force as a short text block, and fine-tuning a video reward model to read it, improves grasp-outcome judgments on object-disjoint held-out data and transfers zero-shot to a different robot, gripper and sensor. The gain over a matched video-only fine-tune is in ranking ($+0.043$ AUROC, $p \approx 0.02$), concentrated on the clips where video is ambiguous, and it disappears when the blocks are shuffled. The representation is the work: the same model with a naive rendering

learned nothing. The approach needs no new architecture and keeps every input inspectable, which makes it a practical way to add contact sensing to the reward models that are starting to label robot data at scale. The next steps are more training clips and a second hand to attack the data limit, calibrated fingertips and more Baxter rollouts to turn the transfer result into a measured one, and higher-resolution video to find out how much of the force gain survives a stronger visual channel. A further step is to evaluate, or supply reward to, a cloud-hosted vision-language-action policy. The `baxter_control` repository already holds a control loop for Dream-1 Dream Labs [2026], a Dream Labs model served through its API: the client sends camera frames, a text prompt and joint state and receives chunks of joint targets, and the service fine-tunes the model on an uploaded demonstration set. Scoring such a policy's rollouts with the tactile reward model, or using the reward to select its fine-tuning data, is the natural next use of this work.

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A Reproducibility

All code, configurations and frozen plans are in the `tactile-flora` repository (branch `hrdexdb-v3-force-representation`). HRDexDB: the pinned source revision, the frozen object-disjoint split, the force configuration, the exported force classifier, the training plan with input hashes and the Slurm job ids are recorded under `docs/experiments/`. Baxter: the firmware, ROS node and recorder are under `tactile_flora/baxter/`; demo collection, dataset publishing and the rollout driver under `baxter/control/`; the ACT training scripts under `baxter/act/`; the rollout source adapter and the preparation and evaluation scripts under `tactile_flora/sources/` and `scripts/cluster/`. Datasets: `OG7407/baxter-cube-pick-left` (demonstrations), `OG7407/baxter-cube-pick-left-rollouts` (rollouts), and `HRDexDB/HRDexDB` at revision `98b8f8e`. Side-by-side videos of every HRDexDB clip the tactile model wins and of the Baxter rollouts, with per-clip numbers, are in the repository's `writeup/` folder.

B Force-only probes on HRDexDB

Measured before training, with object-grouped five-fold cross-validation on the 350 training clips (majority class 68.6%): absolute force levels and contact timing give 74.0% accuracy and AUROC

0.71; trial-relative hold statistics 76 to 77% and 0.78 to 0.80; the implemented summary line on the 64-frame grid 78.3% and 0.78. Only 2 of 110 training failures lack contact and 100 have a contact plateau of at least 5 s, so contact timing alone cannot separate outcomes. Within an object, successful trials hold with more force (true in 31 of 36 training objects with both outcomes) and sustain the hold (late-over-early hold ratio 1.02 against 0.83). The exported classifier has out-of-fold accuracy 78.3% and AUROC 0.764 on the training clips.